

TIB Model

TIB Theory

A unifying construction of orthonormal bases

The general structure of our transfer function is defined as:

$$(0.0.1) \quad H(z) = \sum_{i=1}^k \theta_i \mathcal{B}_i(z)$$

where $\mathcal{B}_i(z)$ is the basis rational function. If we choose a set of inappropriate $\{\mathcal{B}_i(z)\}$, the estimate might be poor. One example is to have:

$$(0.0.2) \quad \mathcal{B}_i(z) = \frac{1}{z - \xi_i}$$

where the poles $\{\xi_i\}$ are chosen according to a *priori* knowledge of the underlying dynamics. Similarly, the Laguerre model is defined as:

$$(0.0.3) \quad \mathcal{B}_i(z) = \left(\frac{\sqrt{1 - \xi^2}}{z - \xi} \right) \left(\frac{1 - \xi z}{z - \xi} \right)^n ; |\xi| < 1$$

and the Kautz model is defined as:

$$(0.0.4) \quad \mathcal{B}_i(z) = \begin{cases} \frac{\sqrt{(1-\alpha^2)(1-\gamma^2)}}{z^2 - \alpha(\gamma+1)z + \gamma} \left(\frac{\gamma z^2 - \alpha(\gamma+1)z + 1}{z^2 - \alpha(\gamma+1)z + \gamma} \right)^{(n-1)/2} \\ \frac{\sqrt{(1-\gamma^2)(z-\alpha)}}{q^2 - \alpha(\gamma+1)z + \gamma} \left(\frac{\gamma z^2 - \alpha(\gamma+1)z + 1}{q^2 - \alpha(\gamma+1)z + \gamma} \right)^{(n/2)} \end{cases}$$

where $|\alpha| < 1$, $|\gamma| < 1$. However, for the models (2), (3), (4) as above, the restriction is that the knowledge of only one pole can be incorporated. Another idea is that,

$$(0.0.5) \quad \mathcal{B}_i(z) = \left(\frac{\sqrt{1 - |\xi_i|^2}}{z - \xi_i} \right) \prod_{k=0}^{n-1} \left(\frac{1 - \bar{\xi}_k z}{z - \xi_k} \right)$$

using this model we allow a variety of poles to be incorporated.

TIB filter

For TIB model we have

$$A = M^{-1}N$$

then the transition equation becomes:

$$Mz_{t+1} = Nz_t + MB\epsilon_t$$

where M and N are lower triangular matrices with bandwidth d (d represents number of inputs). To advance the state vector, we first compute

$$r_t \equiv Nz_t + MB\epsilon_t$$

and then solve the banded matrix equation

$$Mz_{t+1} = r_t$$

For single input case

$$M \equiv \begin{bmatrix} 1 & & & & \\ \gamma_1 & 1 & & & \\ & \gamma_2 & 1 & & \\ & & \gamma_3 & 1 & \\ & & & \ddots & \ddots \end{bmatrix}$$

$$N \equiv \begin{bmatrix} \lambda_1 & & & & \\ \mu_1 & \lambda_2 & & & \\ & \mu_2 & \lambda_3 & & \\ & & \mu_3 & \lambda_4 & \\ & & & \ddots & \ddots \end{bmatrix}$$

where $\rho_k = \sqrt{1 - |\lambda_k|^2}$, $\mu_k = \rho_{k+1}/\rho_k$ and $\gamma_k = \lambda_k^* \mu_k$.

MIMO

pass

Optimality Criteria

H_2 criterion is:

$$\sum_{k=0}^{\infty} |h_k - \hat{h}_k|^2$$

while H_{∞} criterion is:

$$\left\| \begin{pmatrix} h_1 & h_2 & h_3 & \cdots \\ h_2 & h_3 & & \\ h_3 & & \ddots & \\ \vdots & & & \ddots \end{pmatrix} - \begin{pmatrix} \hat{h}_1 & \hat{h}_2 & \hat{h}_3 & \cdots \\ \hat{h}_2 & \hat{h}_3 & & \\ \hat{h}_3 & & \ddots & \\ \vdots & & & \ddots \end{pmatrix} \right\|_2$$

Realization Theory

Let $F(\lambda)$ be a $p \times q$ rational *mvf* that is proper. Then there exists an integer $n > 0$ and matrices $D \in \mathbb{C}^{p \times q}$, $C \in \mathbb{C}^{p \times n}$, $A \in \mathbb{C}^{n \times n}$ and $B \in \mathbb{C}^{n \times q}$ such that

$$F(\lambda) = D + C(\lambda I_n - A)^{-1} B$$

Functions of Toeplitz Matrices

Given a stable and minimum phase $m \times m$ Matrix Valued transfer function

$$f(z) = \sum_{i=0}^{\infty} f_i z^i$$

We denote its corresponding block Toeplitz matrix as:

$$T_f = \begin{bmatrix} f_0 & 0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 & 0 \\ f_2 & f_1 & f_0 & 0 & 0 \\ f_3 & f_2 & f_1 & f_0 & 0 \\ \dots & \dots & & & \end{bmatrix}$$

The input-output mapping relationship is given by:

$$Y(t) = T_f X(t)$$

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ \vdots \end{pmatrix} = \begin{pmatrix} f_0 & 0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 & 0 \\ f_2 & f_1 & f_0 & 0 & 0 \\ f_3 & f_2 & f_1 & f_0 & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} x_t \\ x_{t-1} \\ x_{t-2} \\ x_{t-3} \\ \dots \end{pmatrix}$$

From the Matlab code we have the following relation:

$$Th = b$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -a_0 & 1 & 0 & 0 & 0 \\ -a_1 & -a_0 & 1 & 0 & 0 \\ -a_2 & -a_1 & -a_0 & 1 & 0 \\ -a_3 & -a_2 & -a_1 & -a_0 & \dots \end{pmatrix} \begin{pmatrix} 1 \\ h_1 \\ h_2 \\ h_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Notice that we can also write the our system using Hankel matrix:

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ \vdots \end{pmatrix} =$$

For MA process, we have

$$\begin{aligned} y_t &= h_0 x_t + h_1 x_{t-1} + h_2 x_{t-2} + \dots \\ y_{t-1} &= h_0 x_{t-1} + h_1 x_{t-2} + h_2 x_{t-3} + h_3 x_{t-4} + \dots \\ y_{t-2} &= h_0 x_{t-2} + h_1 x_{t-3} + h_2 x_{t-4} + h_3 x_{t-5} + \dots \end{aligned}$$

then

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \end{pmatrix} = \begin{pmatrix} h_0 & h_1 & h_2 & \dots & \dots \\ 0 & h_0 & h_1 & \dots & \dots \\ 0 & 0 & h_0 & h_1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} x_t \\ x_{t-1} \\ x_{t-2} \\ \vdots \end{pmatrix}$$

Similarly, for AR process, we have

$$\begin{aligned} y_t &= a_1 y_{t-1} + a_2 y_{t-2} + a_3 y_{t-3} + \dots + x_t \\ y_{t-1} &= a_1 y_{t-2} + a_2 y_{t-3} + a_3 y_{t-4} + \dots + x_{t-1} \\ y_{t-2} &= a_1 y_{t-3} + a_2 y_{t-4} + a_3 y_{t-5} + \dots + x_{t-2} \end{aligned}$$

$$\begin{pmatrix} x_t \\ x_{t-1} \\ x_{t-2} \\ \vdots \end{pmatrix} = \begin{pmatrix} 1 & -a_1 & -a_2 & \cdots & \cdots \\ 0 & 1 & -a_1 & -a_2 & \cdots \\ 0 & 0 & 1 & -a_1 & -a_2 & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \vdots \end{pmatrix} \begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \end{pmatrix}$$

then we would have the relationship,

$$\begin{pmatrix} h_0 & h_1 & h_2 & \cdots & \cdots \\ 0 & h_0 & h_1 & \cdots & \cdots \\ 0 & 0 & h_0 & h_1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} 1 & -a_1 & -a_2 & \cdots & \cdots \\ 0 & 1 & -a_1 & -a_2 & \cdots \\ 0 & 0 & 1 & -a_1 & -a_2 & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \vdots \end{pmatrix} = \mathbb{I}_n$$

So we have

$$\begin{pmatrix} h_0 & h_1 & h_2 & \cdots \end{pmatrix} \begin{pmatrix} 1 & -a_1 & -a_2 & \cdots & \cdots \\ 0 & 1 & -a_1 & -a_2 & \cdots \\ 0 & 0 & 1 & -a_1 & -a_2 & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \vdots \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \cdots \end{pmatrix}$$

Taking transpose on both sides, we have:

$$\begin{pmatrix} 1 & & & & \\ -a_1 & 1 & & & \\ -a_2 & -a_1 & 1 & & \\ & -a_2 & -a_1 & 1 & \cdots \\ \vdots & \vdots & & & \cdots \end{pmatrix} \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}$$